

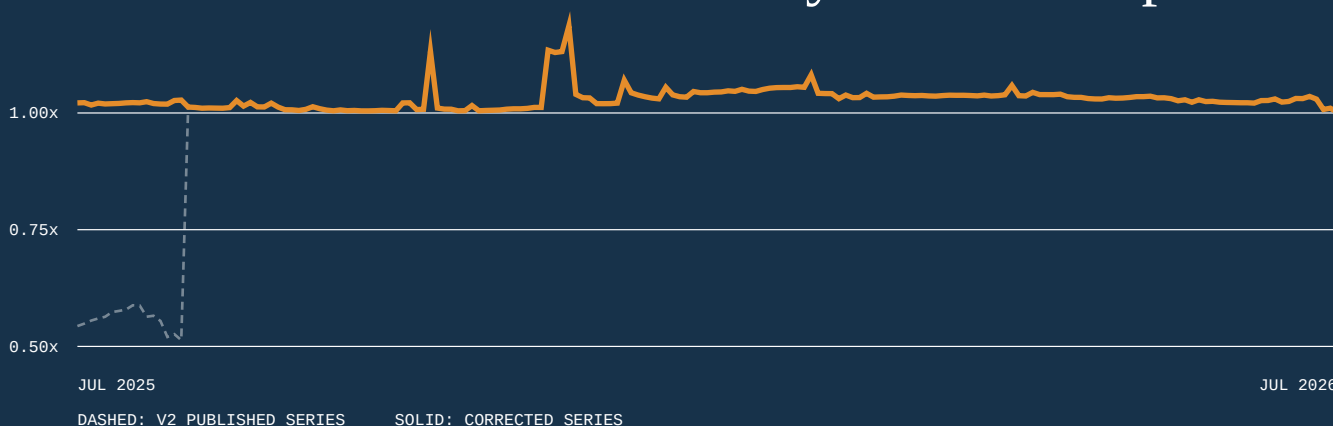
STABLECOIN SOLVENCY / EVIDENCE STATUS: RESEARCH

Accounting for an Omnichain Dollar

A 12-month public-state reconciliation of USDT0, twice corrected.

PUBLISHED RESULT

The documented universe now reconciles to 1.0003. That measured difference is only three basis points.



TSOLMONDORJ NATSAGDORJ / SUWAPPU RESEARCH

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SUWAPPU.BOT/RESEARCH

THE FINDING

Measured completely, the system is indistinguishable from exactly 1:1.

At 01:53 UTC on 1 August 2026, the Ethereum lockbox held \$3.4536B of USDT against \$3.4526B of directly measured USDT0 liabilities. That is a 1.0003 ratio and a \$1.03M measured difference - about three basis points. Because the reads are not block-aligned and cannot see encumbrance or in-flight messages, we do not call that difference a cushion.

\$3.4536B

USDT IN ETHEREUM LOCKBOX

1.0003

MEASURED COLLATERAL / LIABILITIES

183

BLOCK-ALIGNED OBSERVATIONS / 12 MONTHS

WHAT CHANGED

Version 1 omitted liabilities and manufactured a surplus. Version 2 fixed the liability side but verified the wrong Polygon collateral address, manufacturing a shortfall. This revision completes both sides. Zero of 183 corrected panel observations fall below par; the endpoint, however, has wound down to measurement noise.

WHAT DID NOT CHANGE

A public balance reconciliation is a necessary solvency check, not a solvency certificate. It cannot establish legal encumbrance of escrowed USDT, net cross-chain messages in flight, or whether the issuer registry itself is complete. At a 3bp margin, each omission can dominate the measured difference.

DECISION FRAME

Use the on-chain check as an always-on reconciliation primitive. Pair it with evidence about registry completeness, message state, and the legal status of collateral before treating the result as proof of full backing.

CORRECTIONS ARE PART OF THE RESULT

Two errors moved the headline in opposite directions.

Neither error required private information to find. Both were failures to define and verify the accounting perimeter before interpreting the ratio.

VERSION	REPORTED ENDPOINT	BUFFER	WHAT WAS WRONG
V1	1.042	\$137.5M	Liability universe truncated
V2	1.0015 panel	\$5.1M	Wrong Polygon collateral address
V3	1.0003 head	\$1.03M	Complete documented universe

ERROR ANATOMY

- 01

Universe truncation

Omit liabilities and an apparent reserve surplus appears. V1 missed \$134.8M of liabilities; completing the universe removed 96% of the reported buffer.
- 02

Account misattribution

Verify the wrong backing address and a healthy system can appear insolvent. The canonical Polygon predicate held \$1.22B-\$1.39B during the pre-migration window; V2's control read \$0.02 at a different address.
- 03

Timestamp misalignment

Compare balances from different moments and migration flow becomes a false mismatch. The panel resolves each chain to a block at the same target timestamp; the event window is rescanned every six hours.

OPERATING RULE

Define the liability universe, verify every account, align time - then interpret the ratio.

ACCOUNTING INVARIANT

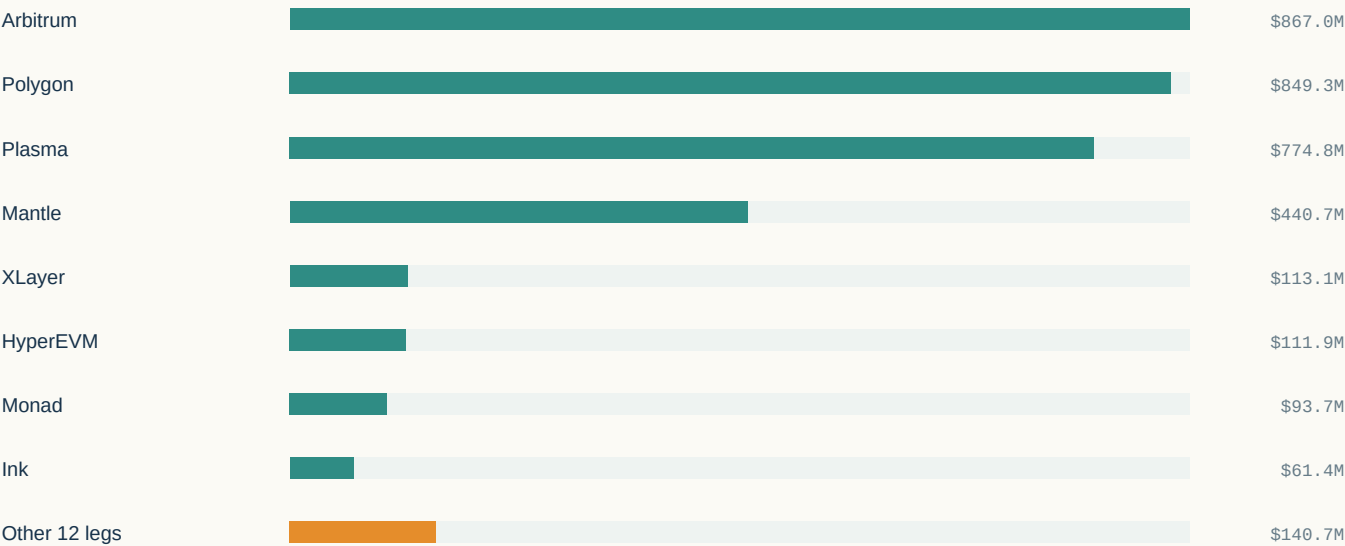
One escrow. Twenty directly measured liability legs.

For a lock-and-mint omnichain dollar, the public-state test is simple: collateral held in the canonical Ethereum lockbox should cover the sum of remote USDT0 supply. HyperCore is verified as a sub-ledger of HyperEVM and is not double-counted.

1 AUG 2026 / 01:53 UTC

\$3.4536B collateral / \$3.4526B liabilities = 1.0003x

DIRECT LIABILITY SNAPSHOT



20 legs directly measured

0 excluded

1 verified sub-ledger

HEAD_SNAPSHOT_20260801.JSON

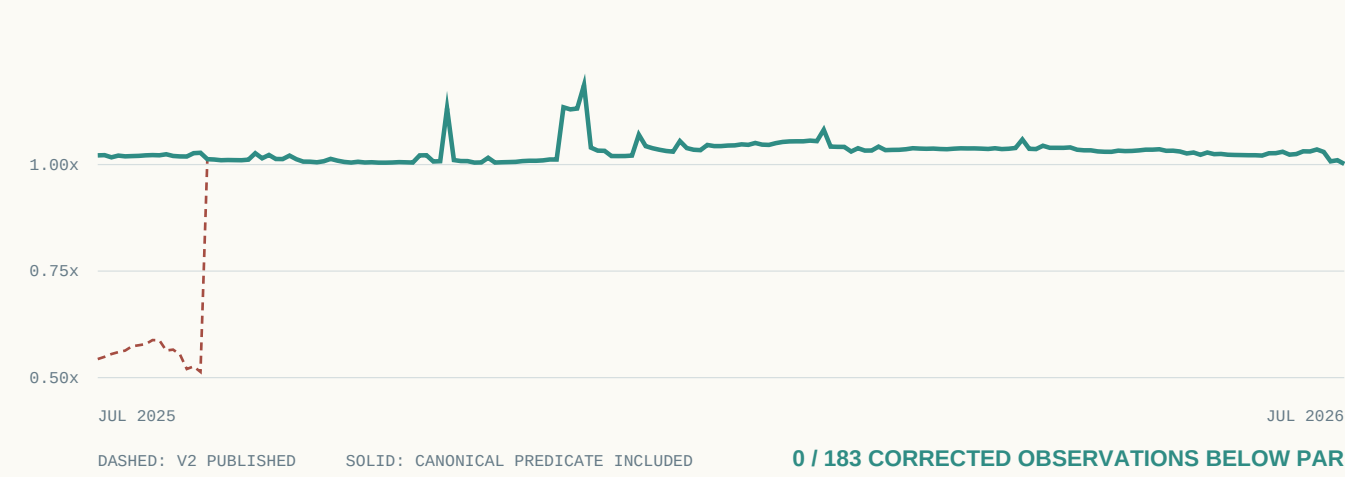
INTERPRETATION

The \$1.03M arithmetic difference is smaller than uncertainties this read cannot observe. We therefore report par, not a positive reserve cushion. The precision belongs to the measurement; the conclusion respects its limits.

CORRECTION 02

The apparent shortfall was a wrong-address artifact.

Before 27 August 2025, Polygon PoS USDT backing sat in the canonical Polygon predicate, not the USDT0 lockbox. V2 checked a different address and interpreted the missing \$1.3B as an unidentified reserve account.



SIX-HOUR MIGRATION BRACKET / 27 AUG 2025

-\$1,358.8M

CANONICAL PREDICATE

+\$1,258.6M

ETHEREUM LOCKBOX

-\$98.1M

ARBITRUM SUPPLY

\$2.1M

RESIDUAL FLOW

INDEPENDENT CROSS-CHECK

The \$1,358.8M predicate outflow matched Polygon supply at bracket open (\$1,358.759M) to within about \$82K - 0.006% of the flow. The issuer had also announced the backing migration that day. Chain state and the public statement describe the same event.

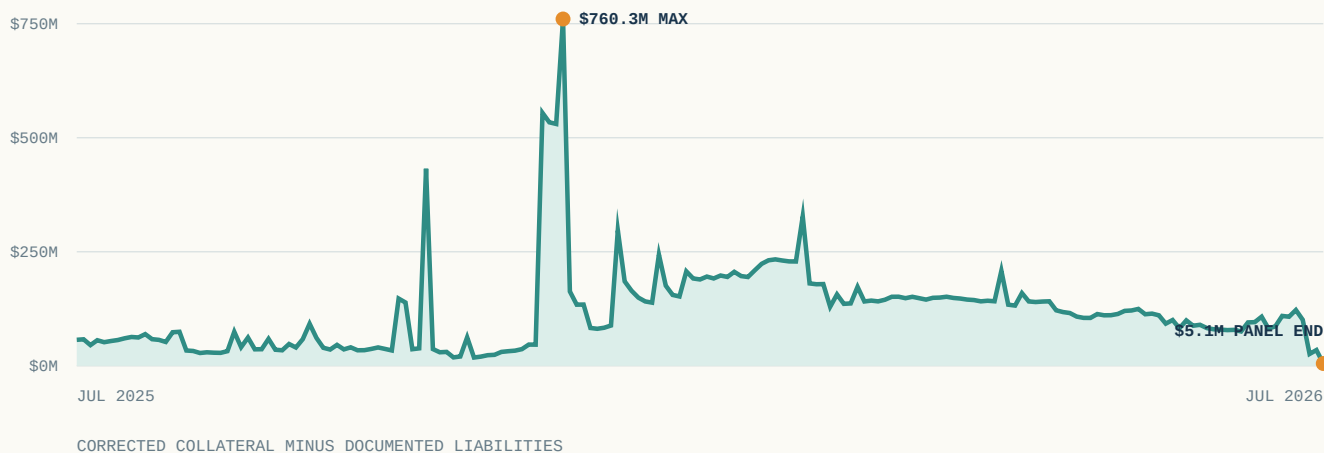
LESSON

A break in a reserve time series can be an accounting-boundary migration, not an economic impairment. Verify the address map before fitting the story.

THE ENDPOINT MATTERS

The historical cushion was wound down toward par.

The measured buffer ranged from roughly 15bp to 18.7% of liabilities across the year - a 124x span inconsistent with a visible proportional target. The end of the sample is more decision-relevant than the historical exceedance.

**\$760.3M**

MAX / 15 DEC 2025

\$296.9M

31 DEC 2025

\$122.4M

17 JUL 2026

\$5.1M

25 JUL / PANEL END

FINAL EIGHT DAYS

-\$192.9M collateral vs -\$75.7M liabilities

About \$117M more collateral left than liabilities declined. The panel buffer fell to \$5.1M; the complete-universe head check six days later measured \$1.03M.

INTERPRETATION

Historically above par does not mean meaningfully overcollateralized now. At the endpoint, measurement uncertainty is the headline.

NECESSARY, NOT SUFFICIENT

What public chain state can - and cannot - prove.

A balance reconciliation answers one narrow question very well. Treating it as a complete reserve attestation would erase exactly the uncertainty that becomes decisive at a three-basis-point margin.

VISIBLE FROM PUBLIC STATE

NOT ESTABLISHED BY THE READ

● Canonical escrow balance

USDT balanceOf() on the verified Ethereum OAdapter lockbox.

● Remote token supply

totalSupply() on each documented USDT0 deployment, with decimals and live contracts checked.

● Accounting perimeter used

Twenty direct liability legs, plus an explicit HyperCore containment check; Tron and TON classified as Legacy Mesh.

● Legal encumbrance

A positive token balance does not establish that the collateral is unencumbered or bankruptcy-remote.

● Messages in flight

A point-in-time sum does not expose net mint/burn instructions moving between chains.

● Registry completeness

The method can verify documented deployments; it cannot prove that the issuer's deployment registry omits nothing.

A MONITORING STACK SHOULD TRACK

01 Complete liability registry

02 Collateral + legacy escrows

03 Net messages in flight

04 Account-level encumbrance evidence

05 Boundary migrations and contract upgrades

06 Corrections to the address map

THRESHOLD

At \$1.03M, any single invisible item above roughly three basis points can flip the sign of the measured buffer.

AUDIT THE RESULT - AND THE MISTAKES

Everything needed to rerun the measurement is public.

The panel uses public RPC endpoints, no explorer API, no indexer, and no credentials. Superseded series are retained so a reader can reproduce the exact errors that produced earlier conclusions.

MEASUREMENT DESIGN

01

183 aligned observations

Every 48 hours from 26 July 2025 through 25 July 2026, with block resolution per chain at each target timestamp.

02

Six-hour event rescan

The 25 Aug-1 Sep migration window is rescanned at six-hour resolution; the canonical predicate is backfilled at all 16 pre-break panel blocks.

03

Complete-universe head check

Twenty direct liability legs measured in one session on 1 Aug 2026, with the HyperCore containment relationship verified and no documented direct leg excluded.

PROOF FILES

`data/usdt0_timeseries.csv`

183-row panel + retained wrong-address control

`data/polygon_predicate_prebreak.json`

canonical Polygon backing backfill

`data/head_snapshot_20260801.json`

complete-universe head measurement

`data/buffer_dynamics.json`

flow coupling + terminal drawdown

`code/collect_usdt0.py`

panel collection harness

`code/head_snapshot.py`

current-universe snapshot

OPEN REPLICATION BUNDLE

suwappu.bot/research/replication

WHAT TO DO WITH THE RESULT

Make the accounting perimeter a product surface.

This paper's most reusable result is operational: a cross-chain asset becomes easier to underwrite when the deployment registry, backing accounts, and boundary changes are published in machine-readable form.

01

For issuers

Publish a dated machine-readable registry of every liability leg, backing account, containment relationship, and migration. A reserve ratio is only as good as the address map behind it.

02

For allocators and routers

Automate the public-state reconciliation, but do not equate it with a full reserve attestation. Escalate when the measured difference approaches the uncertainty in messages, encumbrance, or registry coverage.

03

For monitor builders

Version the accounting perimeter itself. Contract migrations, new deployments, sub-ledgers, and address corrections are state changes that should be observable and historically replayable.

DISCLOSURES

Research, not investment advice. Suwappu builds cross-chain execution infrastructure and holds operational stablecoin balances, including USDT and USDT0, incidental to running it. No directional position informed this analysis. Tether, Everdawn Labs, and other named parties did not review the work before publication. All inputs are public chain state or cited public documents. The working paper is canonical where this report and the paper differ.

SELECTED SOURCES

USDT0 technical documentation: deployments and Legacy Mesh (accessed 31 Jul 2026)

USDT0, Polygon USDT Now Upgraded to USDT0 (27 Aug 2025)

Polygon PoS Bridge documentation: canonical ERC20 predicate (accessed 31 Jul 2026)

Everdawn Labs: USDT0 audit reports; Chaos Labs: USDT0 Mechanism Design Review (2025)

Newey-West (1987); Politis-Romano (1994); Bai-Perron (1998)

[READ THE WORKING PAPER + INSPECT THE EVIDENCE](#)[SUWAPPU.BOT/RESEARCH/REPLICATION](https://suwappu.bot/research/replication)